Comparator

Comparator Latch – Design Source File

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-- Group 17
-- Senevirathne S.M.P.U.
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-- Create Date: 04/13/2024 10:02:06 PM
-- Design Name: Comparator
-- Module Name: Comparator_Latch - Behavioral
-- Project Name: Nanoprocessor
-- Target Devices: Basys3 Board

----------------------------------------------------------------------------------


library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

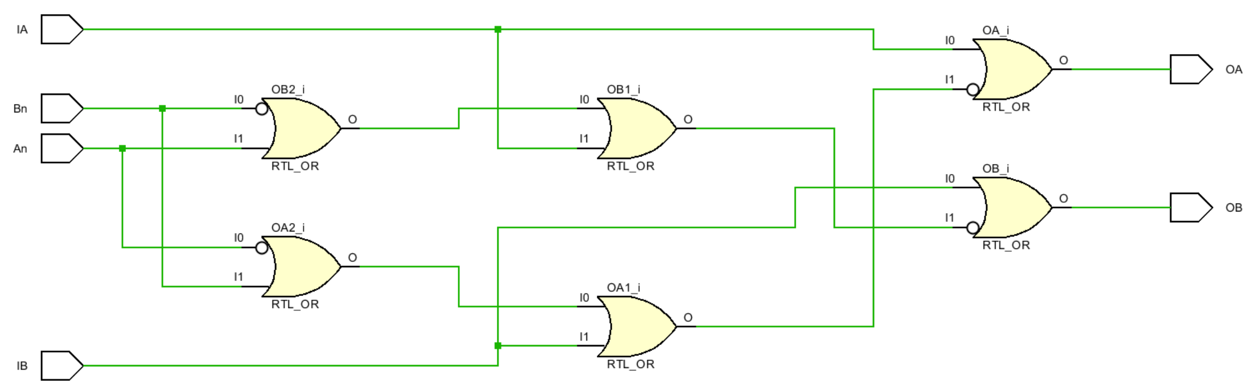
entity Comparator_Latch is
  Port ( An : in STD_LOGIC; --n th bit of number A
       Bn : in STD_LOGIC; --n th bit of number B
       IA : in STD_LOGIC; --input status: A
       IB : in STD_LOGIC; --input status: B
```

```
        OA : out STD_LOGIC; --output status: A
        OB : out STD_LOGIC); --output status: B
end Comparator_Latch;

architecture Behavioral of Comparator_Latch is

begin
   OA <= IA OR NOT(NOT(An) OR Bn OR IB); -- output status: A
   OB <= IB OR NOT(NOT(Bn) OR An OR IA); -- output status: B
end Behavioral;
```

Comparator Latch – Schematic Diagram



Comparator – Design Source File

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-- Create Date: 04/13/2024 10:02:06 PM

-- Design Name: Comparator

-- Module Name: Comparator - Behavioral

-- Project Name: Nanoprocessor

-- Target Devices: Basys3 Board

----------------------------------------------------------------------------------

library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity Comparator is

  Port ( EN : in STD_LOGIC; --Enable input

      A : in STD_LOGIC_VECTOR (3 downto 0); --4 bit value: A

      B : in STD_LOGIC_VECTOR (3 downto 0); --4 bit value: B

      AGreat : out STD_LOGIC; --Is A the greater number?

      BGreat : out STD_LOGIC; --Is B the greater number?

      Equal : out STD_LOGIC); --Are the numbers equal?

end Comparator;

architecture Behavioral of Comparator is

component Comparator_Latch

  Port ( An : in STD_LOGIC; --n th bit of number A

    Bn : in STD_LOGIC; --n th bit of number B

    IA : in STD_LOGIC; --input status: A

    IB : in STD_LOGIC; --input status: B

    OA : out STD_LOGIC; --output status: A

    OB : out STD_LOGIC); --output status: B
```
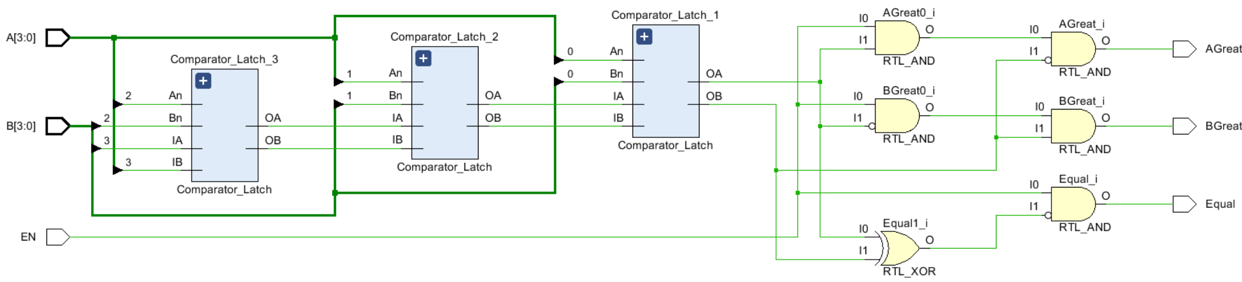

```
end component;

SIGNAL IA3, IA2, IA1, IB3, IB2, IB1 : std_logic; --signal input statuses of A and B
SIGNAL OA3, OA2, OA1, OB3, OB2, OB1 : std_logic; --signal output statuses of A and B
SIGNAL An2, An1, An0, Bn2, Bn1, Bn0 : std_logic; --signal 2 downto 0 bits of A and B
begin
  Comparator_Latch_3 : Comparator_Latch --mapping the bottom latch
  port map (
    An => An2,
    Bn => Bn2,
    IA => IA3,
    IB => IB3,
    OA => OA3,
    OB => OB3
  );
  Comparator_Latch_2 : Comparator_Latch --mapping the middle latch
  port map (
    An => An1,
    Bn => Bn1,
    IA => IA2,
    IB => IB2,
    OA => OA2,
    OB => OB2
  );
  Comparator_Latch_1 : Comparator_Latch --mapping the top latch
  port map (
    An => An0,
    Bn => Bn0,
    IA => IA1,
```

```
    IB => IB1,
    OA => OA1,
    OB => OB1
  );
--Assigning the inputs to the bottom latch
IA3 <= B(3);
IB3 <= A(3);
An2 <= A(2);
Bn2 <= B(2);

--Assigning the inputs to the middle latch
IA2 <= OA3;
IB2 <= OB3;
An1 <= A(1);
Bn1 <= B(1);

--Assigning the inputs to the top latch
IA1 <= OA2;
IB1 <= OB2;
An0 <= A(0);
Bn0 <= B(0);

--Defining the flags; AGreat, BGreat and Equal
AGreat <= EN AND OA1 AND NOT(OB1);
BGreat <= EN AND NOT(OA1) AND OB1;
Equal <= EN AND NOT(OA1 XOR OB1);

end Behavioral;
```

Comparator – Schematic Diagram



Comparator – Simulation File

-- Group 17

-- Senevirathne S.M.P.U.

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-- Create Date: 04/13/2024 10:02:06 PM

-- Design Name: Comparator

-- Module Name: Comparator_TB - Behavioral

-- Project Name: Nanoprocessor

-- Target Devices: Basys3 Board

----

library IEEE;

use IEEE.STD_LOGIC_1164.ALL;

entity Comparator_TB is

end Comparator_TB;

```
architecture Behavioral of Comparator_TB is
  component Comparator is
    Port ( EN : in STD_LOGIC; --Enable input
         A : in STD_LOGIC_VECTOR (3 downto 0); --4 bit value: A
         B : in STD_LOGIC_VECTOR (3 downto 0); --4 bit value: B
         AGreat : out STD_LOGIC; --Is A the greater number?
         BGreat : out STD_LOGIC; --Is B the greater number?
         Equal : out STD_LOGIC); --Are the numbers equal?
  end component;

  --Signal EN, A, B, AGreat, BGreat and Equal
  SIGNAL EN : std_logic;
  SIGNAL A : std_logic_vector (3 downto 0);
  SIGNAL B : std_logic_vector (3 downto 0);
  SIGNAL AGreat : std_logic;
  SIGNAL BGreat : std_logic;
  SIGNAL Equal : std_logic;


begin
  Comparator_Sim : Comparator --mapping comparator for simulation
    port map(
      EN => EN,
      A => A,
      B => B,
      AGreat => AGreat,
      BGreat => BGreat,
      Equal => Equal);
```

```
process begin
    wait for 100ns;
    EN <= '1';

    A <= "0000"; -- A = 0

    B <= "0000"; -- B = 0, Equal
    wait for 100ns;

    B <= "0111"; -- B = 7, BGreater
    wait for 100ns;

    B <= "1000"; -- B = -8, AGreater
    wait for 100ns;

    EN <= '0';

    A <= "0111"; -- A =7

    B <= "0000"; -- B = 0, AGreater
    wait for 100ns;

    B <= "0111"; -- B = 7, Equal
    wait for 100ns;

    B <= "1000"; -- B = -8, AGreater
    wait for 100ns;
```

```
EN <= '1';

A <= "1000"; -- A = -8

B <= "0000"; -- B = 0, BGreater
wait for 100ns;

B <= "0111"; -- B = 7, BGreater
wait for 100ns;

B <= "1000"; -- B = -8, Equal
wait for 100ns;

EN <= '0';

A <= "0000"; -- A = 0

B <= "0000"; -- B = 0, Equal
wait for 100ns;

B <= "0111"; -- B = 7, BGreater
wait for 100ns;

B <= "1000"; -- B = -8, AGreater
wait for 100ns;

EN <= '1';

A <= "0111"; -- A =7
```

```
        B <= "0000"; -- B = 0, AGreater
        wait for 100ns;

        B <= "0111"; -- B = 7, Equal
        wait for 100ns;

        B <= "1000"; -- B = -8, AGreater
        wait for 100ns;

        EN <= '0';

        A <= "1000"; -- A = -8

        B <= "0000"; -- B = 0, BGreater
        wait for 100ns;

        B <= "0111"; -- B = 7, BGreater
        wait for 100ns;

        B <= "1000"; -- B = -8, Equal
        wait for 100ns;

        wait;
    end process;

end Behavioral;
```

Comparator – Timing Diagram